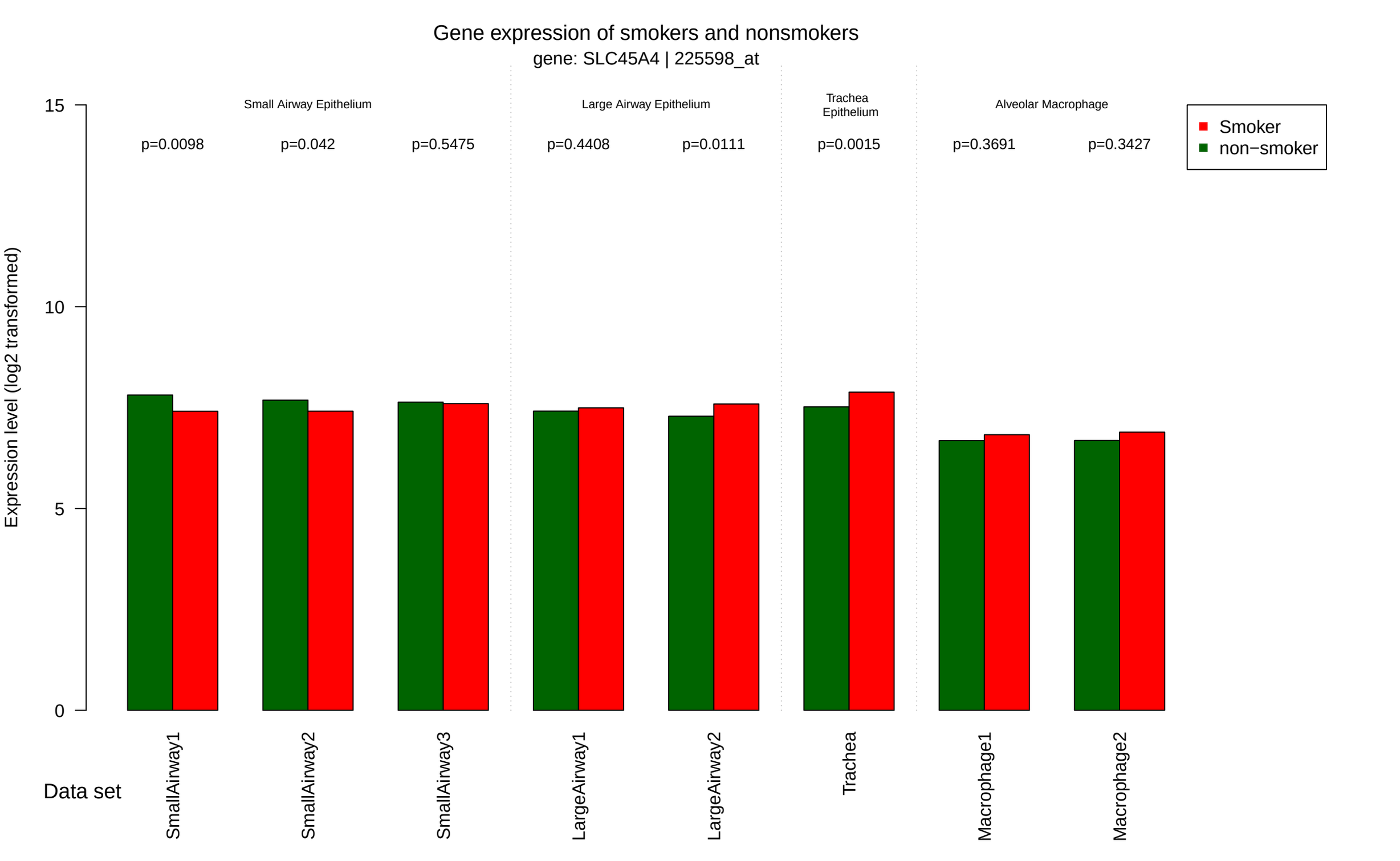
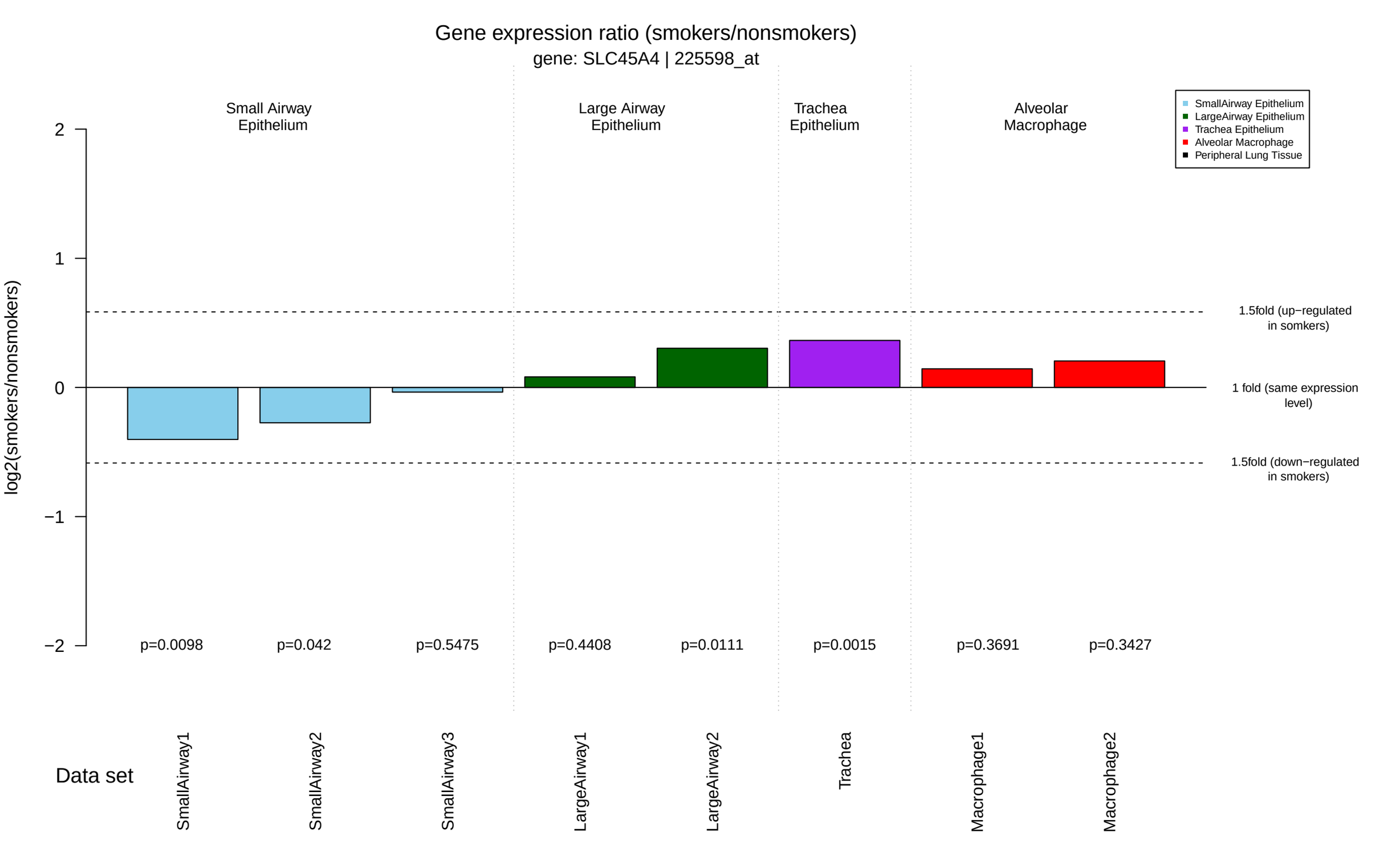
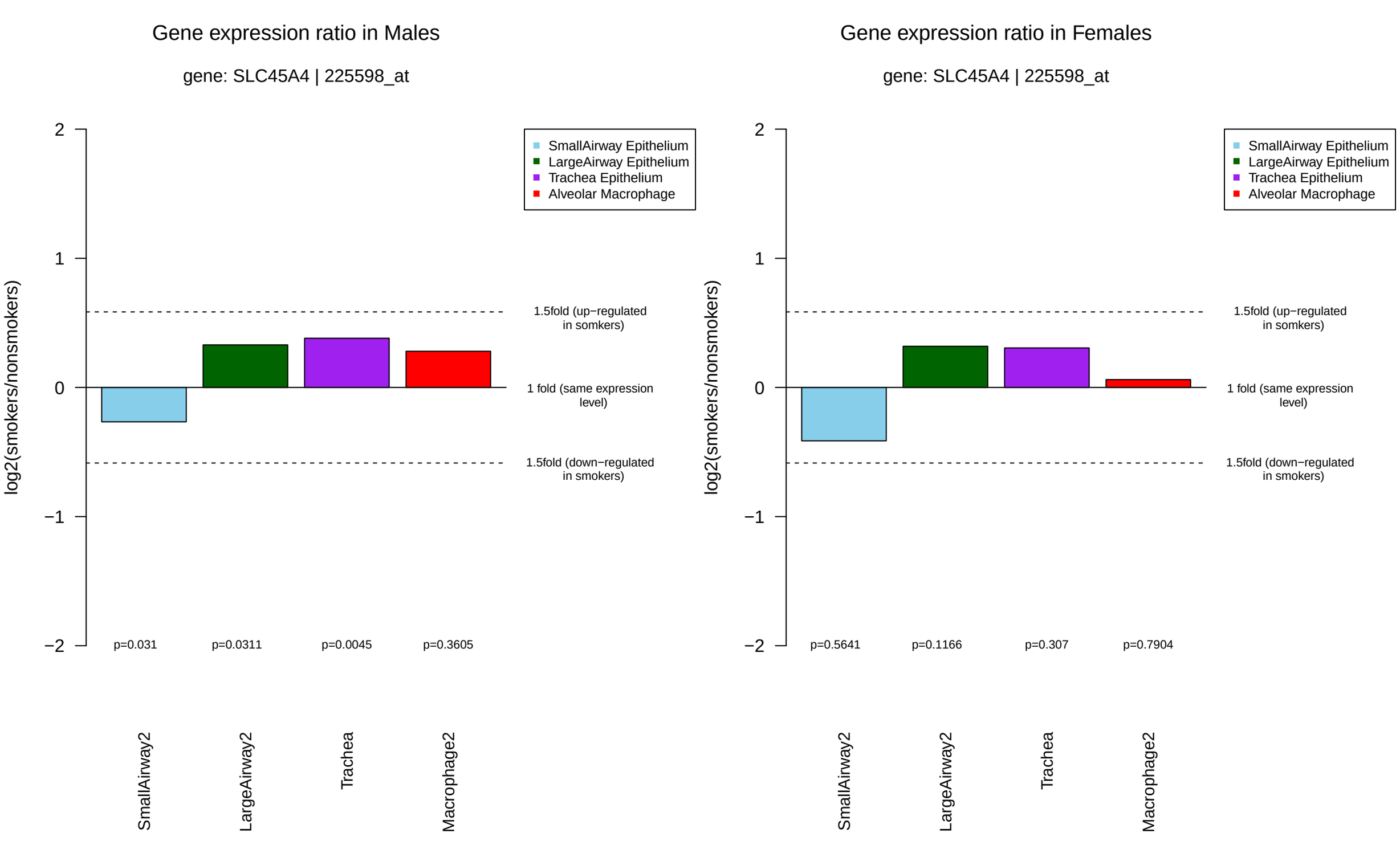
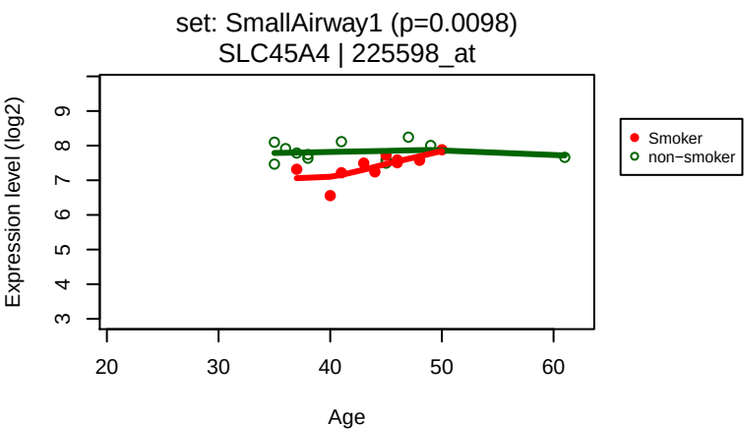
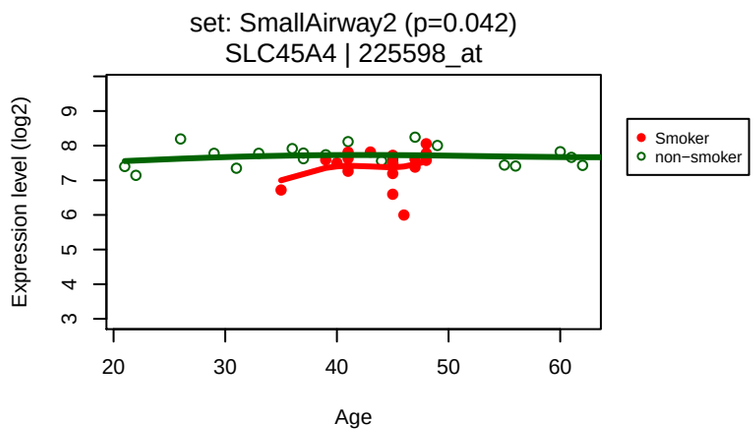
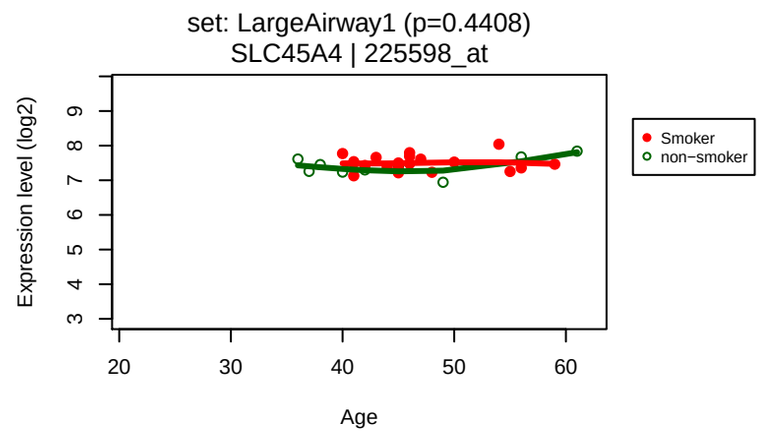

Gene expression of smokers and nonsmokers
gene: SLC45A4 | 225598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0098
p=0.042
p=0.5475
p=0.4408
p=0.0111
p=0.0015
p=0.3691
p=0.3427
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

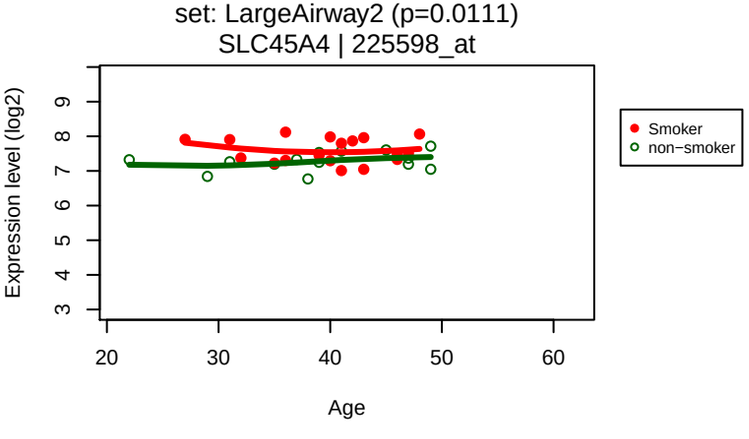

Gene expression ratio (smokers/nonsmokers)
gene: SLC45A4 | 225598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0098
p=0.042
p=0.5475
p=0.4408
p=0.0111
p=0.0015
p=0.3691
p=0.3427
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

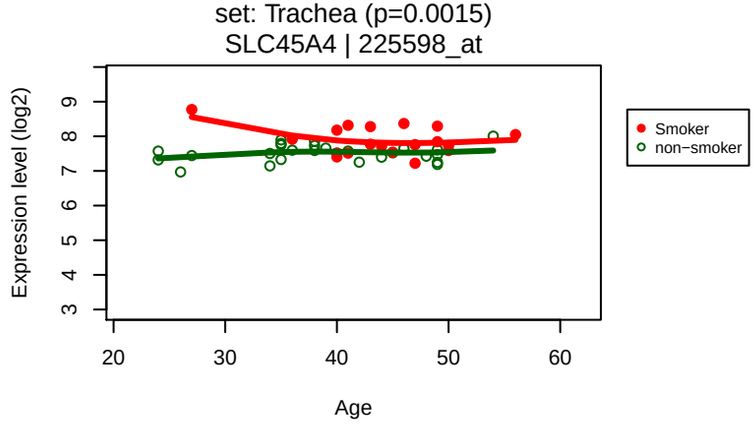

Gene expression ratio in Males
gene: SLC45A4 | 225598_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.031
p=0.0311
p=0.0045
p=0.3605
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC45A4 | 225598_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5641
p=0.1166
p=0.307
p=0.7904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

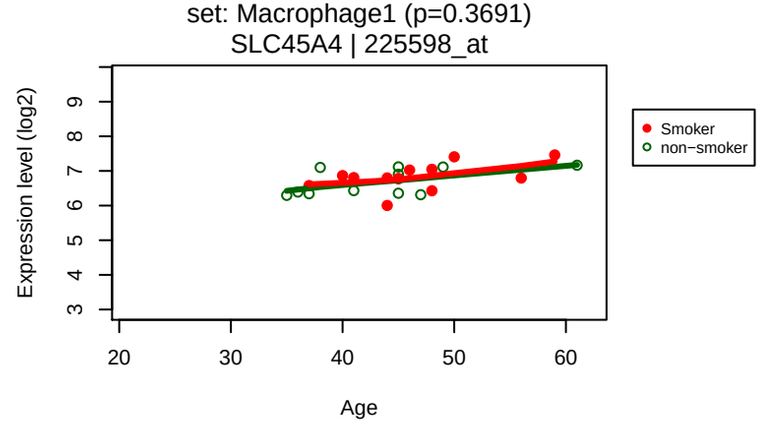

set: SmallAirway1 (p=0.0098)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker

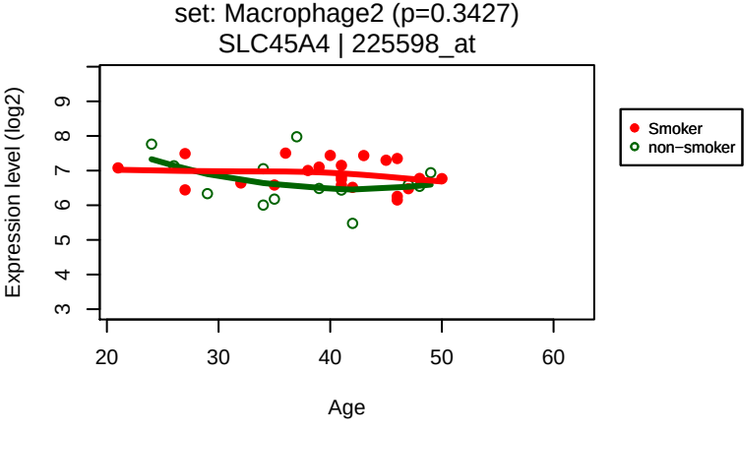

set: SmallAirway2 (p=0.042)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker

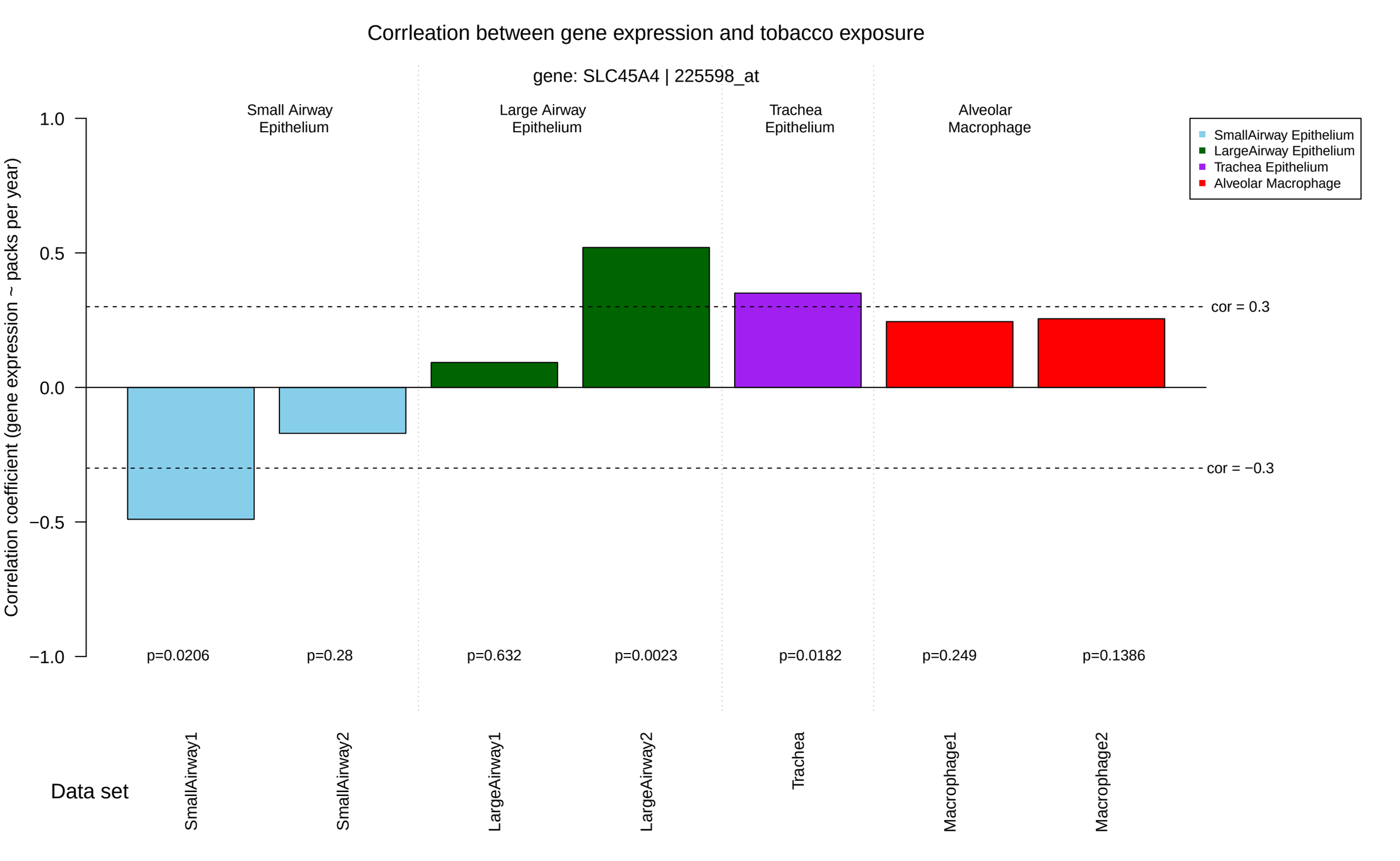

set: LargeAirway1 (p=0.4408)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0111)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0015)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3691)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3427)
SLC45A4 | 225598_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC45A4 | 225598_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0206
p=0.28
p=0.632
p=0.0023
p=0.0182
p=0.249
p=0.1386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2